C PROGRAMMING :

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <semaphore.h>
#include <sys/mman.h>
#include <sys/wait.h>
int main(){
sem_t *sem;
sem = mmap(NULL, sizeof(sem_t),PROT_READ | PROT_WRITE,MAP_SHARED | MAP_ANONYMOUS,-1, 0);
sem_init(sem, 1, 1);
if (fork() == 0){
sem_wait(sem);
printf("Child Process Entering Critical Section\n");
sleep(3);
printf("Child Process Leaving Critical Section\n");
sem_post(sem);
exit(0);}
sem_wait(sem);
printf("Parent Process Entering Critical Section\n");
sleep(3);
printf("Parent Process Leaving Critical Section\n");
sem_post(sem);
wait(NULL);
sem_destroy(sem);
return 0;}
```

OUTPUT :

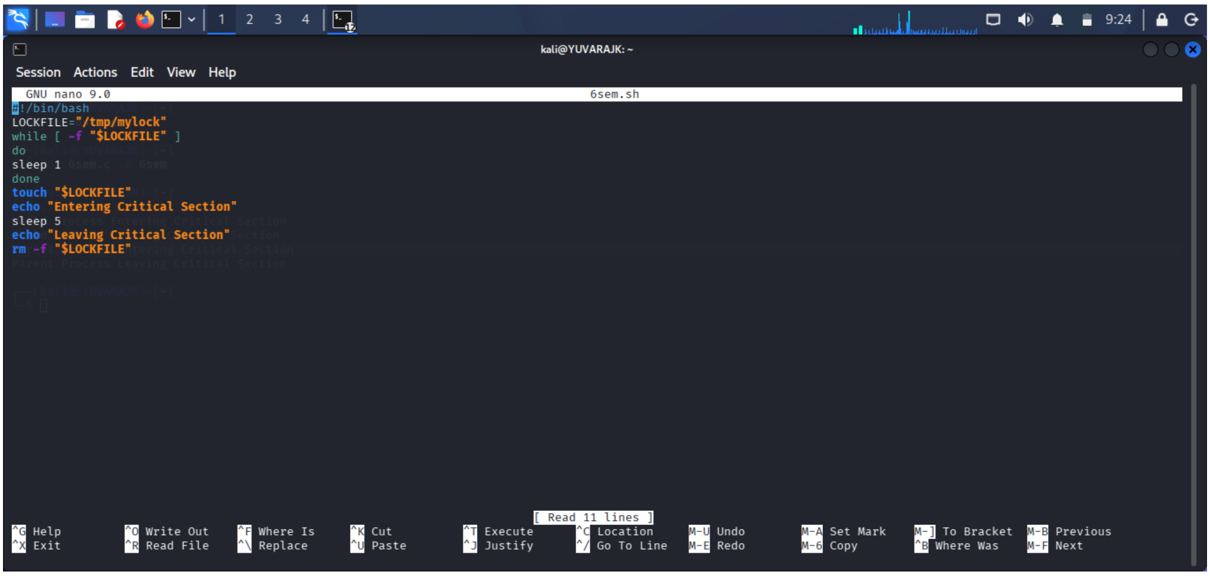
```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿YUVARAJK)-[~]
└─$ nano 6sem.c

┌──(kali㉿YUVARAJK)-[~]
└─$ gcc 6sem.c -o 6sem

┌──(kali㉿YUVARAJK)-[~]
└─$ ./6sem
Child Process Entering Critical Section
Child Process Leaving Critical Section
Parent Process Entering Critical Section
Parent Process Leaving Critical Section

┌──(kali㉿YUVARAJK)-[~]
└─$
```

SHELL PROGRAMMING :



OUTPUT :